// Implement the diffie Hellman Key Exchange mechanism using HTML and JavaScript. //Consider the end user as one of the parties (Alice) and the JavaScript application as //other party (Bob)

```
<!DOCTYPE html>
<html>
<body>
<script>

// This program calculates the Key for two persons
// using the Diffie-Hellman Key exchange algorithm

// Power function to return value of a ^ b mod P
function power(a, b, p)
{
   if (b == 1)
      return a;
   else
      return((Math.pow(a, b)) % p);
}

// Driver code
var P, G, x, a, y, b, ka, kb;

// Both the persons will be agreed upon the
// public keys G and P
 var P = window.prompt("Enter value of P: ");
// A prime number P is taken
//P = 23;
document.write("The value of P:" + P + "<br>");
 var G = window.prompt("Enter value of G: ");
// A primitive root for P, G is taken
//G = 9;
document.write("The value of G:" + G + "<br>");

// Alice will choose the private key a
// a is the chosen private key
//a = 4;
var a = window.prompt("Enter private key for Alice: ");
document.write("The private key a for Alice:" +
            a + "<br>");

// Gets the generated key
x = power(G, a, P);

// Bob will choose the private key b
// b is the chosen private key
//b = 3;
var b = window.prompt("Enter private key for bob: ");
document.write("The private key b for Bob:" +
            b + "<br>");
```

```
// Gets the generated key
y = power(G, b, P);

// Generating the secret key after the exchange
// of keys
ka = power(y, a, P); // Secret key for Alice
kb = power(x, b, P); // Secret key for Bob

document.write("Secret key for the Alice is:" +
            ka + "<br>");
document.write("Secret key for the Bob is:" +
            kb + "<br>");

// This code is contributed by Ankita saini

</script>


</body>
</html>
```